Roll No.: DL.AI.U4AID25009

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Amrita School of Artificial Intelligence, Faridabad

B.Tech Mid-Term Examinations – October 2025

First Semester

Artificial Intelligence & Data Science

23AID102 Elements of Computing - 1

Duration: Two hours Maximum: 50 Marks

Course Outcomes (COs):

CO	Course Outcomes
CO1	Demonstrate proficiency in performing number system conversion, manipulating Boolean Algebra expressions and realization of basic gates, Combinational circuit.
CO2	Demonstrate proficiency in designing Sequential logic, Flip Flops, Registers, Counters. Von-Neumann architecture, Arithmetic logic unit (ALU).
CO3	Understand instruction set architecture and computer system components.

Answer all questions

Question No-1

Q. 1A. Convert the decimal number 213 into 8-bit binary. [1][CO1][BTL 2]

Q. 1B. Find the 1's and 2's complement of 10101101. [1][CO1][BTL 2]

Q. 1C. State De Morgan's Theorems. [1][CO1][BTL 2]

Q. 1D. Write the Sum of Products (SOP) form for:
$F(A,B,C) = (A + B')(A' + C)$ [1][CO1][BTL 2]

Q. 1E. Draw the logic symbol of a Half Adder and write its Boolean expression?[1][CO1][BTL2]

Question No-2. Perform following arithmetic operations using 2's complement and 8-bit sign representation:
(a) $-28 + 32$
(b) $-58 - 67$
[6][CO1][BTL2]

Question No-3. Simplify the Boolean expression using algebraic rules and draw logic diagram:
$F = A'B'C + A'BC + AB'C + ABC + AB'C'$ [6][CO1][BTL2]

Question No-4. Draw the Full Adder truth table, derive Sum and Carry expressions, and implement logic diagram using XOR, AND, OR gates. [6][CO1][BTL2]

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Question No-5. Design a 8-to-1 multiplexer using logic gates to implement the Boolean function:

$$F(A,B,C) = \Sigma(1,5,6)$$

Use A, B and C as the selection inputs and show how the data inputs (D0–D7) are assigned. [6][CO1][BTL2]

Question No-6. Solve using K-map with don't care conditions: Simplify $F(A,B,C,D) = \Sigma(0,2,5,7,8,10,13,15)$, with $D(1,3,9,11)$ as don't cares. Draw minimal SOP circuit. [7][CO1][BTL3].

Question No-7. Implement the function $F(A,B,C) = \Sigma(1,3,5,7)$
(a) in SOP form,
(b) using only NAND gates,
(c) verify the output using truth table.

[7][CO1][BTL3].

Question No-8. Design a 4-bit binary adder using full adder circuits to add two 4-bit numbers, $A = A_3A_2A_1A_0$ and $B = B_3B_2B_1B_0$.

Tasks:

1. Draw the block diagram showing how four full adders are connected.
2. Indicate the carry propagation between stages.
3. Label the sum outputs $S_3S_2S_1S_0$ and the final carry-out C_4.
4. Write the truth table for the addition of $A = 1010$ and $B = 0011$.
5. Show the final binary sum and the decimal equivalent. [7][CO1][BTL3].

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Course Outcome /Bloom's Taxonomy Level (BTL) Mark Distribution Table

CO	Marks	BTL	Marks
CO1	50	BTL 2	29
		BTL 3	21

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